# complaint-router

Lab 2: data pipeline, versioning, baseline and MLflow

KSE MLOps · Lab 2 · Zakhar Staryk · github.com/goodfrog33/kse-mlops-complaint-router (tag lab-2)

**39**
monthly batches through Airflow

**0.633 vs 0.380**
macro-F1, model vs rule

**0.0000**
reproduce difference

# At a glance

## Data versioned

- 312,000 complaints, 39 months, DVC
- MinIO remote + public read-only mirror

## Pipeline

- Airflow 3.3.1, six tasks, one month per run
- Reruns leave identical rows

## Quality gate

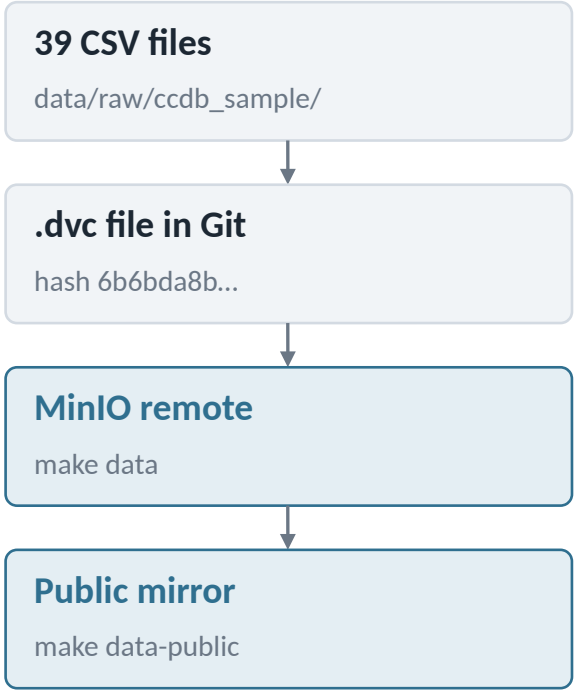
- Fails on purpose (fault injection)
- Fails for real: 2025-12 → 2026-06

## Model and lineage

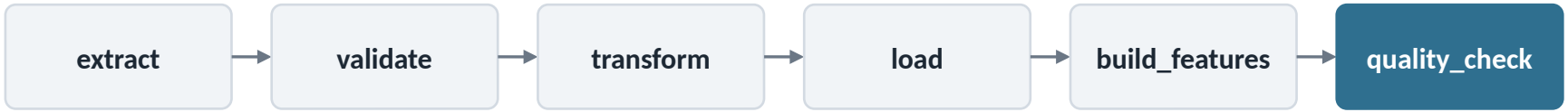
- TF-IDF ensemble 0.633 vs keyword rule 0.380
- MLflow v4 @champion, reproduced exactly

# Raw data is versioned with DVC

| | |
|---|---|
| **Sample** | 39 monthly CSVs × 8,000 complaints (168 MB) |
| **Drawn by** | hash of complaint_id: uniform, byte-identical reruns |
| **Data version** | `6b6bda8b4c8ece9715a5c9127a47dc8e` |
| **Remotes** | MinIO s3://dvc · public HTTP mirror |
| **Pull** | make data · make data-public |
| **Checksum** | `872e979f783b101ff87505a5a74b5838` |



# One Airflow run per month, never the clock




Batch

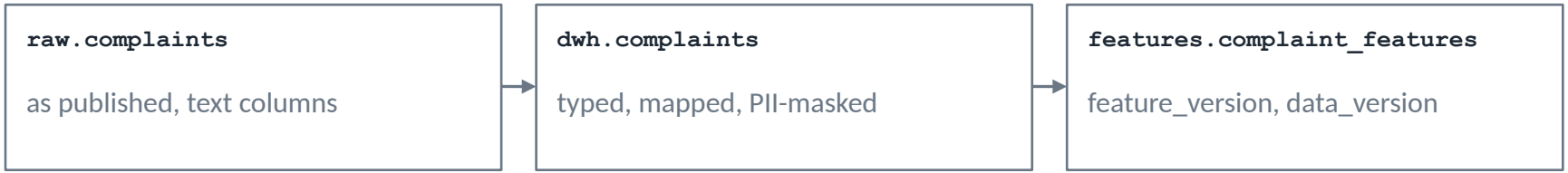
`data_interval_start`

Schedule (@monthly)

`CronDataIntervalTimetable`

Concurrency

**6 months at once: 39 in 174 s**




PostgreSQL warehouse

# Rerunning a month leaves identical rows

| Run 1 · 2024-06 | rows |
|---|---|
| raw.complaints | **8000** |
| dwh.complaints | **8000** |
| features | **2683** |
| checks passed | **5 / 5** |

| Run 2 · 2024-06 | rows |
|---|---|
| raw.complaints | **8000** |
| dwh.complaints | **8000** |
| features | **2683** |
| checks passed | **5 / 5** |

Content hash, before and after

`30048bcc…`

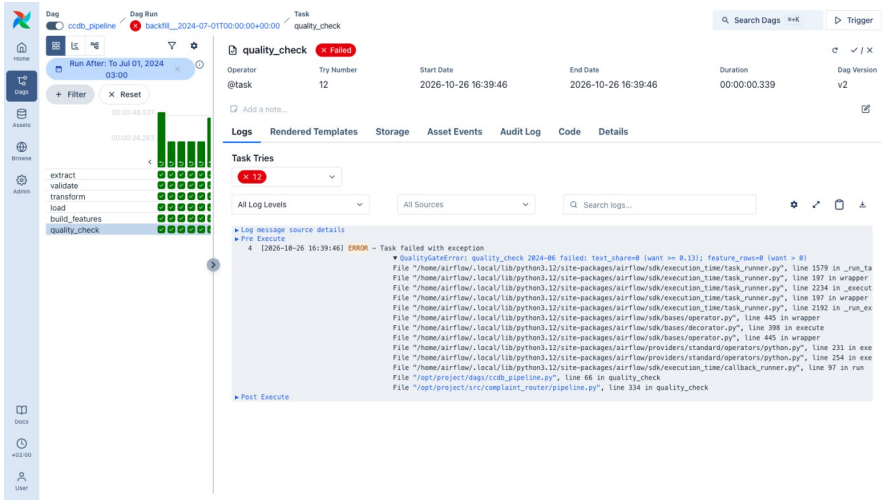
**IDENTICAL**

md5 of every feature row of the month

**Why: every load deletes the month, then inserts it, in one transaction**

```
delete from <table> where batch_month = %s;  copy <table> from stdin
```

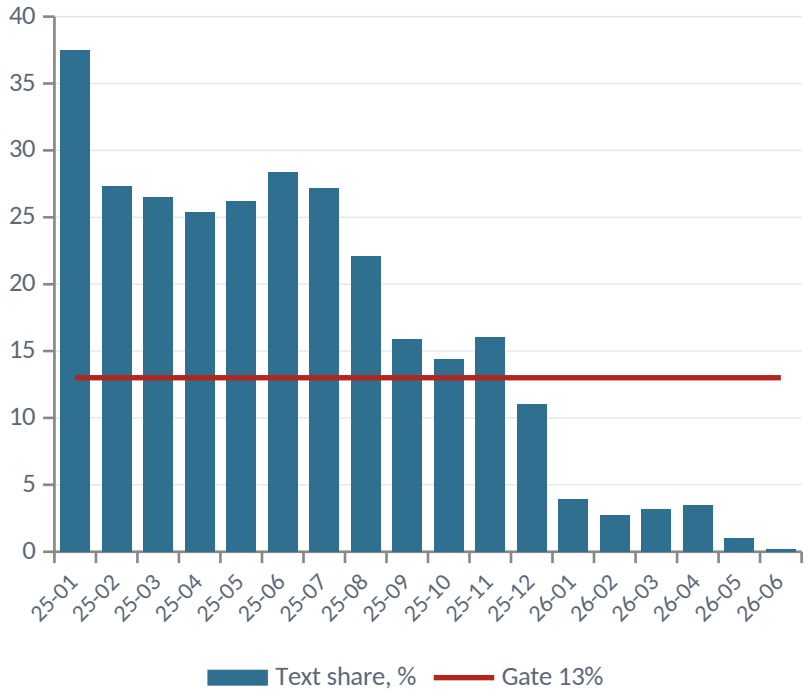
# The quality gate fails when it should




**On purpose: FAULT=null_narratives**

```
text_share=0 (want >= 0.13);
feature_rows=0 (want > 0)
```

For real: text share per month vs gate



# Every feature has a serving-time source

| Feature | Training | Serving |
|---|---|---|
| text | dwh narrative, clean_text_expr | request, clean_text (tested equal) |
| submitted_via | CFPB field | request; missing → unknown |
| state | CFPB field | request; missing → unknown |
| **company** | normalized name | dropped: sources differ |

## Feast: point-in-time join

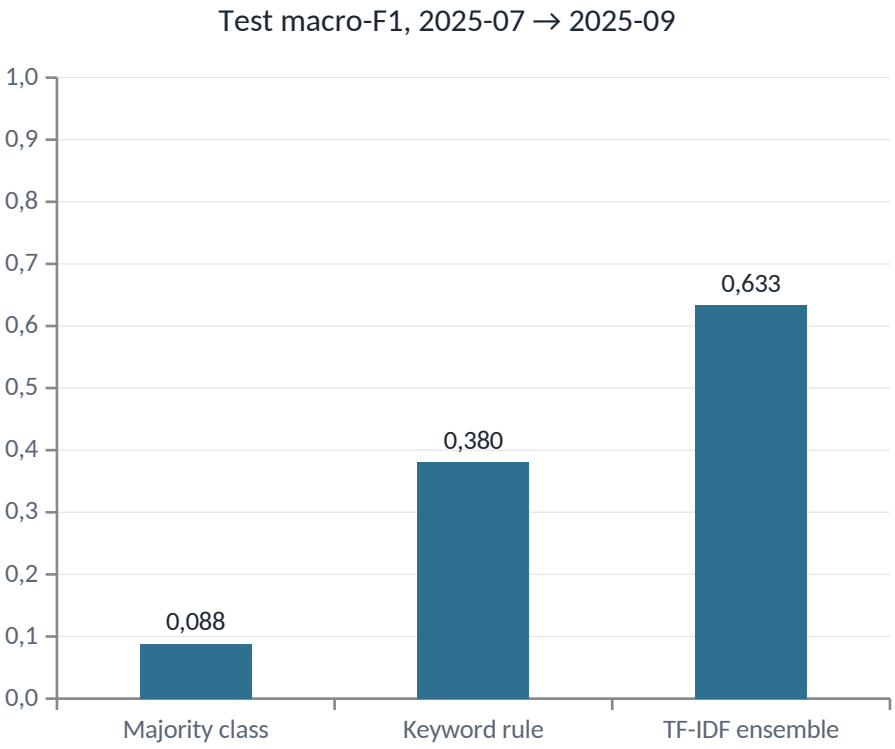
Complaint on 2024-07-15; feature rows at 07-01 and 08-01

| Company | Got | Later |
|---|---|---|
| Acima Credit | 5 | 1 |
| Ally Financial | 7 | 13 |
| American Express | 22 | 23 |

**EARLIER ROW EVERY TIME**

```
make feast-demo
```

# The model beats the rule on every split

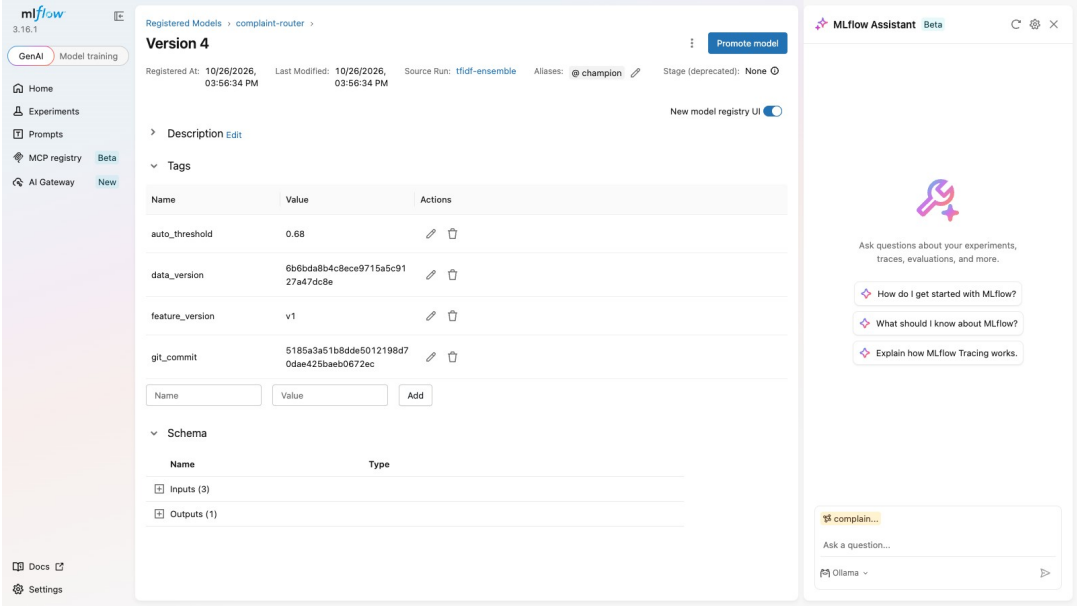



| Backtest month | Model | Rule |
|---|---|---|
| 2025-07 | 0.615 | 0.386 |
| 2025-08 | 0.714 | 0.381 |
| 2025-09 | 0.618 | 0.375 |
| **mean ± std** | **0.649 ± 0.046** | 0.381 ± 0.004 |

**Random split: 0.688**

+0.055 too optimistic: it trains on months after the test. Time split matches use

# Every run carries its data version and commit




Server

**MLflow 3.16.1 · Postgres backend · MinIO artifacts**

Registered

**complaint-router v4 · @champion**

Lineage tags

**data_version · git_commit**

Rule baselines, backtest folds and the random-split check are logged with the same tags.

# One command reproduces the champion

1. Read tags from registry → 2. git worktree at commit → 3. dvc pull + hash check → 4. Rebuild features, retrain

```
$ make reproduce
version 4: commit 5185a3a51b, data 6b6bda8b…, test_macro_f1 0.6325
registered 0.6325  reproduced 0.6325  |diff| 0.0000  tolerance 0.005  -> OK
```

**0.0000**

difference on the same platform

**0.005**

tolerance: BLAS and thread differences

# Lab 2 rows of the maturity scorecard

| Property | Evidence |
|---|---|
| **Data versioned** | make data-public restores 39 files, checksum 872e979f… |
| **Pipeline idempotent** | 2024-06 twice: 8000 / 8000 / 2683, same hash |
| **Quality gate present** | fault fails quality_check; 2025-12 → 2026-06 fail for real |
| **Experiments tracked** | every run: params, metrics, data_version, git_commit |
| **Model registered with lineage** | v4 @champion; make reproduce → diff 0.0000 |

### Run it

```
make setup
make up
make data-seed
make data
make backfill
make train
make reproduce
```

CI: lint, unit and pipeline tests

Repository: [github.com/goodfrog33/kse-mlops-complaint-router/tree/lab-2](github.com/goodfrog33/kse-mlops-complaint-router/tree/lab-2)